US ECONOMICS ANALYST

The Long-Run Effects of Technological Disruption on Workers

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- As concerns about the labor market impact of AI grow, we draw on four decades of individual-level data to assess the short- and long-term "scarring" effects of being displaced by technology. We identify occupations disrupted by technology in each decade since 1980 and use the National Longitudinal Surveys—which follow over 20,000 individuals born in the 1950–60s and 1980s over their entire lives—to track the labor market outcomes of workers employed in these disrupted occupations.
- We draw four main conclusions. First, workers displaced from technology-disrupted occupations face more difficult short-run transitions back into employment. They take approximately one month longer to find a new job and suffer real earnings losses of more than 3% upon reemployment, compared with negligible losses for workers displaced from more stable occupations. A key mechanism behind these worse outcomes is occupational downgrading. Workers displaced by technology are more likely to move into more routine occupations requiring fewer analytical and interpersonal skills, likely because the same technological shifts that eliminated their positions also eroded the value of their existing skills.
- Second, the effects of technological displacement are long-lasting. Over the ten years following a job loss, real earnings for technology-displaced workers grow nearly 10pp less than for never-displaced workers and 5pp less than for other displaced workers, and their probability of experiencing another unemployment spell remains persistently higher. The scarring effects also spill over into broader economic outcomes. Focusing on workers displaced early in their careers, we find that technological displacement slows wealth accumulation—largely through delayed homeownership—and delays household formation.
- Third, the impact of technological displacement varies considerably across workers. Younger, college-educated, and urban workers experience cumulative real earnings losses only half as large as those of other technology-displaced workers. Workers with shorter tenure at the time of displacement also fare better, likely because their skills are less occupation-specific and therefore more transferable. Retraining also helps to cushion the impact of displacement. Workers who participate in retraining programs after displacement experience smaller labor market losses, as retraining facilitates transitions up the occupational ladder into jobs requiring more analytical skills that complement

information and communication technology.

- Fourth, recessions amplify the costs of technological displacement. Firms disproportionately shed routine jobs during downturns, when they are under greater pressure to identify efficiencies. For workers, technology-driven displacement in recessions widens the gaps already observed relative to other displaced workers in expansions—by about three weeks of additional unemployment and 5pp each for the probability of subsequent unemployment and labor force exit—likely because reemployment becomes harder and job matches less durable when many displaced workers compete for limited vacancies at once.
- Overall, these patterns suggest that AI-driven displacement could impose lasting costs on affected workers, with substantially larger effects when job losses coincide with a recession. Contrary to current concerns that the costs of AI will fall especially hard on new graduates, younger workers have actually been able to adjust more flexibly through occupational mobility and skill upgrading in the past.

The Long-Run Effects of Technological Disruption on Workers

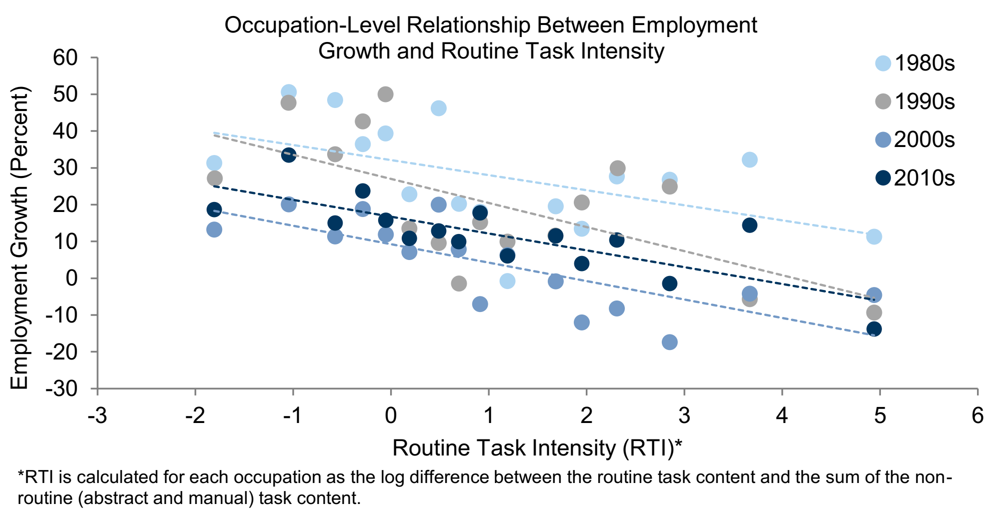
The rapid advance of artificial intelligence (AI) has deepened concerns about its potential to disrupt—or even eliminate—entire occupations. Our Global Economics team estimated that AI could displace 6–7% of US workers over the next decade, raising the unemployment rate by up to ½pp relative to trend during a ten-year transition period, with larger unemployment rate increases if the adoption timeline is compressed.

Many workers will benefit from AI-driven increases in productivity and real income. But how costly is technological disruption for workers who are displaced? In this week's *Analyst*, we draw on four decades of individual-level data from the National Longitudinal Surveys to assess the short- and long-term "scarring" effects of technological disruption on US workers.

Identifying Technology-Disrupted Occupations

We identify technology-disrupted occupations by analyzing employment trends across more than 300 occupation categories in the American Community Survey since the 1980s. Occupations with high routine task intensity—a measure developed by economist David Autor and co-authors[1] to capture the degree to which jobs involve routine, codifiable tasks rather than abstract cognitive work or non-routine manual dexterity—experienced significantly weaker employment growth across recent decades, consistent with their greater exposure to automation (Exhibit 1).

Exhibit 1: Since the 1980s, Employment Growth Has Been Lower in Routine Occupations...

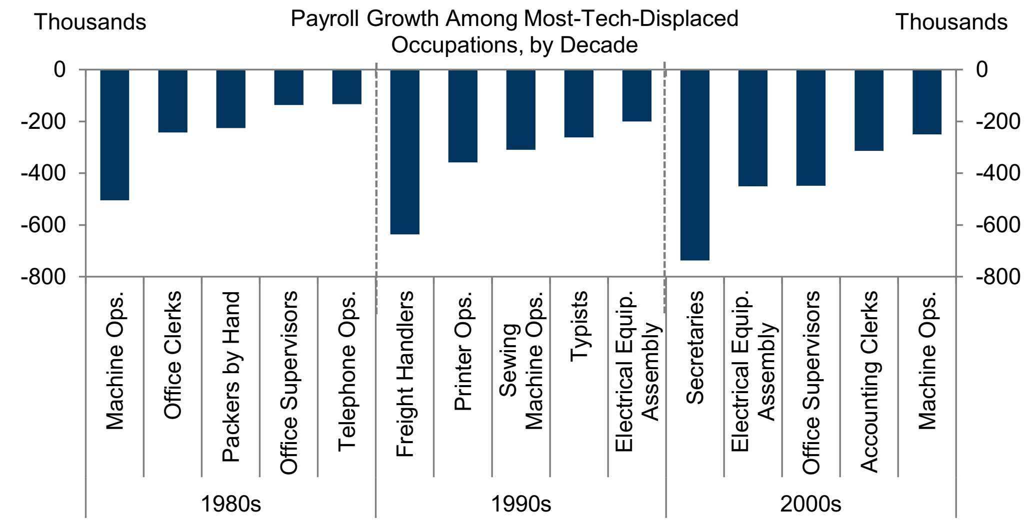


Source: Goldman Sachs Global Investment Research, Census Bureau, Autor and Dorn (2013)

In each decade, we define as technology-disrupted those occupations that fall in the bottom quartile of employment growth and, among those, in the top half of the routine task intensity distribution. Exhibit 2 shows the five technology-disrupted occupations with the largest employment declines over the first three decades of our sample.

[1] Autor, Levy and Murnane (2003), "The Skill Content of Recent Technological Change: An Empirical Exploration," *Quarterly Journal of Economics* and Autor and Dorn (2013), "The Growth of Low-Skill Service Jobs and the Polarization of the US Labor Market," *American Economic Review*.

Exhibit 2: ...Which Are More Easily Automatable and Therefore More Exposed to Technological Disruption



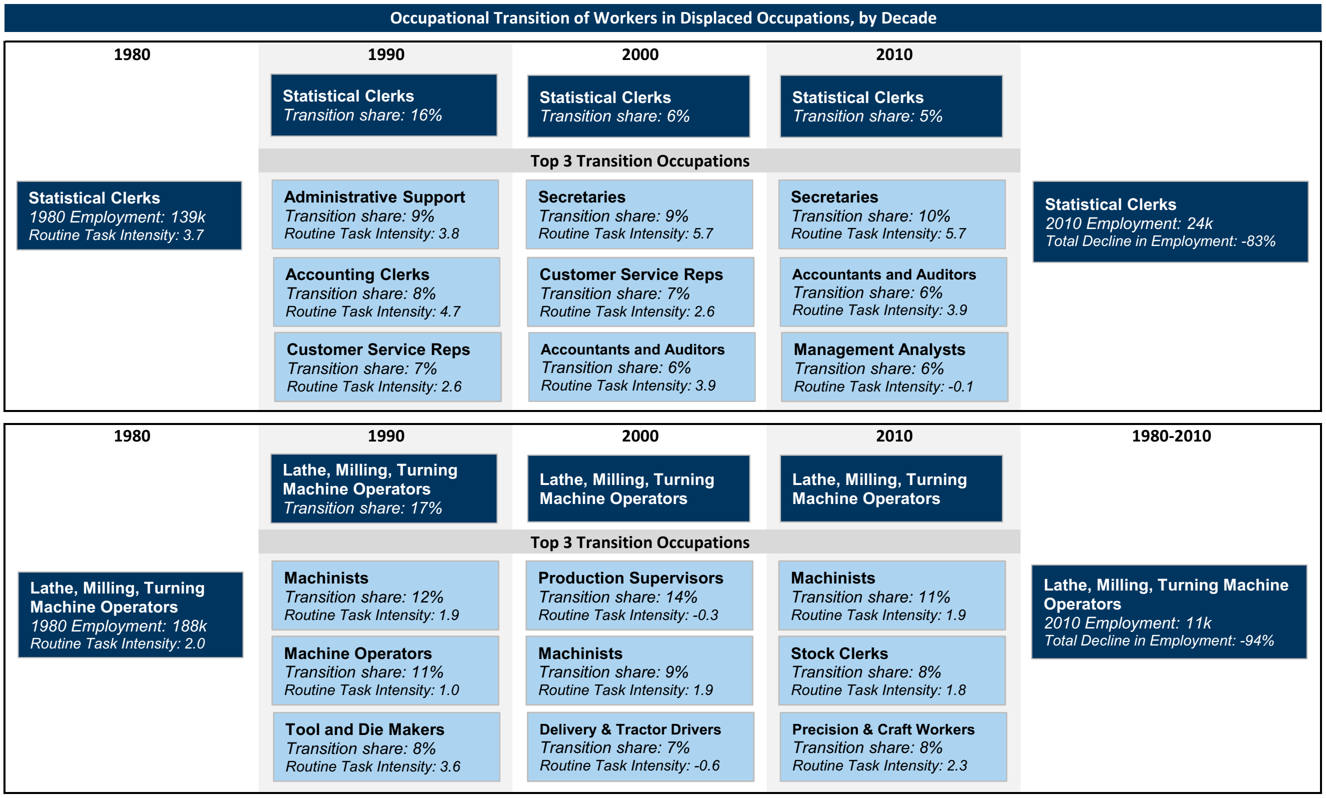
Source: Goldman Sachs Global Investment Research, Census Bureau, Autor and Dorn (2013)

We then study the labor market outcomes of workers in these technology-disrupted occupations using individual-level data from the National Longitudinal Surveys. These nationally representative surveys, sponsored by the Bureau of Labor Statistics (BLS), follow over 20,000 US individuals comprising two cohorts—the first born in the 1950-1960s and followed from 1979 to 2022, and the second born in the 1980s and followed from 1997 to 2021.[2]

This dataset allows us to track workers' full occupational histories. As an illustration, Exhibit 3 shows the occupations into which workers employed as statistical clerks or machine operators in the 1980s—two roles heavily disrupted by technology—transitioned over subsequent decades. The patterns suggest that these workers typically moved into roles requiring comparable skills and carrying, on average, a roughly similar degree of exposure to technological disruption.

[2] The 1979-2022 cohort is tracked annually through 1994 and biennially thereafter, while the 1997-2021 cohort is tracked annually through 2011 and biennially thereafter.

Exhibit 3: Technology-Displaced Workers Typically Transitioned into Roles Requiring Comparable Skills and Carrying Similar Exposure to Technological Disruption



Note: Transition share denotes the share of workers employed as statistical clerks and lathe, milling, turning machine operators in 1980 that transitioned into the listed occupation in each subsequent decade.

Source: Goldman Sachs Global Investment Research, Census Bureau, US Bureau of Labor Statistics

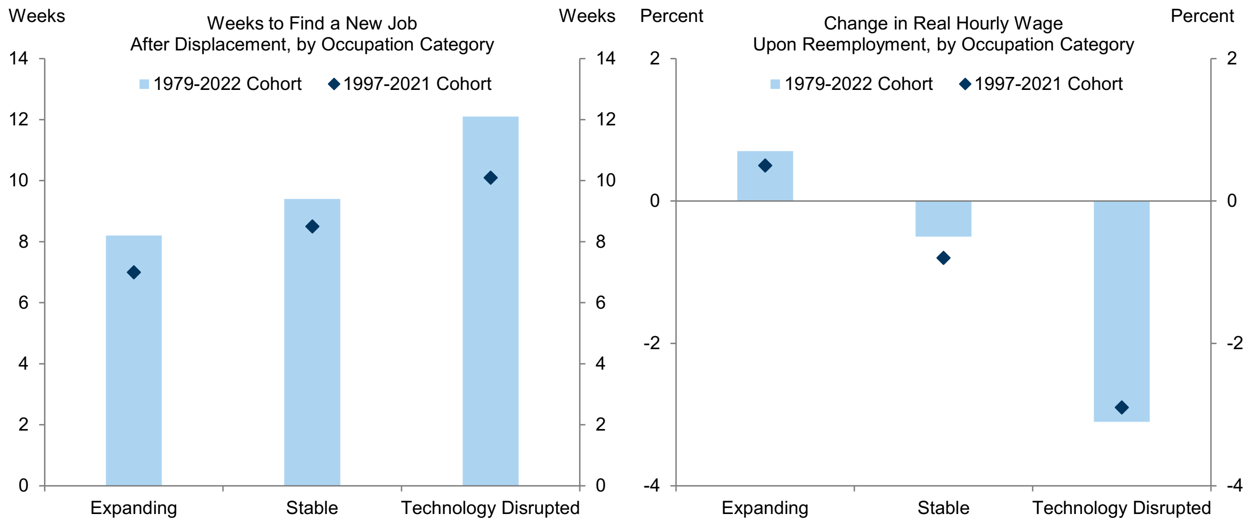
We analyze the career paths of workers displaced by technology throughout our sample and draw four main conclusions.

1. Workers Displaced by Technology Face More Difficult Labor Market Transitions

Workers displaced from technology-disrupted occupations face more difficult transitions back into employment after losing their jobs. Throughout our analysis, we benchmark their outcomes against workers in "stable" occupations—those in the middle two quartiles of employment growth—and "expanding" occupations—those in the top quartile. Relative to these benchmarks, technology-displaced workers take roughly one month longer to find a new job and suffer real earnings losses 3pp larger upon reemployment when moving between full-time positions (Exhibit 4).

These estimates are broadly stable across the two cohorts in our sample and closely aligned with our previous analysis of technological displacement over 1994-2024 using the Current Population Survey's Displaced Worker Supplement.

Exhibit 4: Technology-Displaced Workers Take Longer to Find New Jobs and Suffer Larger Real Earnings Losses Upon Reemployment

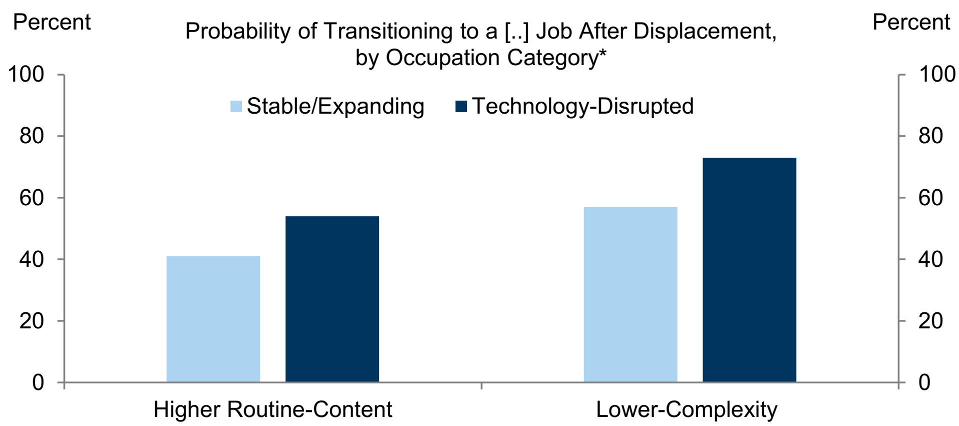


Source: Goldman Sachs Global Investment Research, Census Bureau, US Bureau of Labor Statistics

A key mechanism driving these worse outcomes is occupational downgrading. To gauge the change in job quality before and after displacement, we assess occupations along two dimensions: their routine task intensity (RTI)—capturing the prevalence of repetitive, rule-based tasks—and their complexity—measured using scores from the BLS's National Compensation Survey that reflect the analytical and interpersonal demands of a given role.

As Exhibit 5 shows, technology-displaced workers are more likely upon reemployment to move into occupations with higher RTI and lower complexity—likely because the same technological changes that eliminated their positions also eroded the value of their existing skills.

Exhibit 5: Technology-Displaced Workers Are More Likely to Move into Routine, Lower-Complexity Roles, Likely Because Their Skills Have Been Eroded by Technology



For each occupation, routine content is measured using the Autor and Dorn (2013) routine task intensity index, while complexity is measured using an aggregate score that captures knowledge requirements, analytical and interpersonal skills from the Bureau of Labor Statistics' National Compensation Survey.

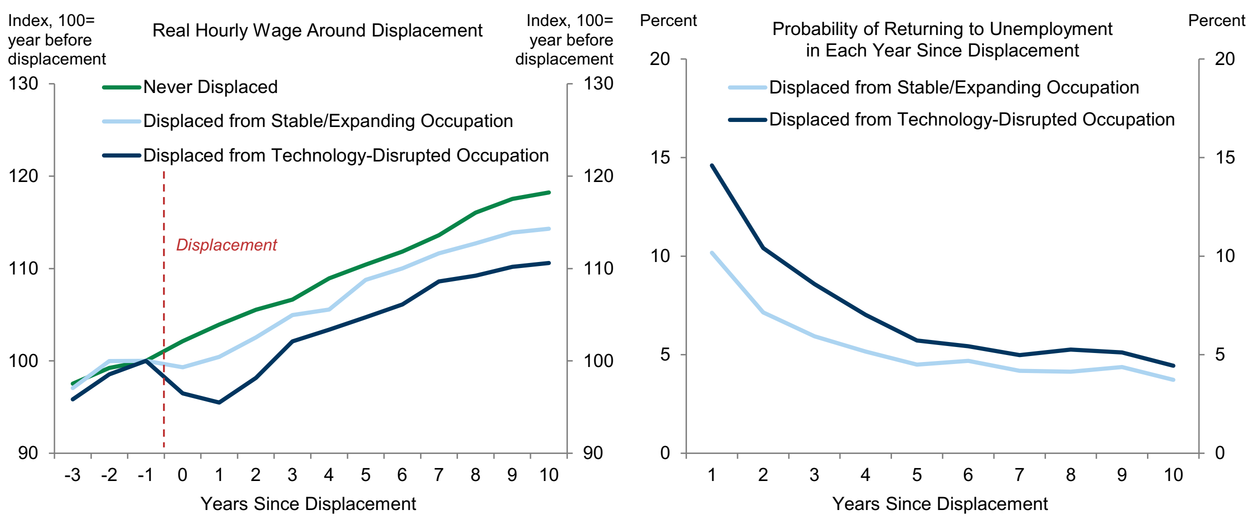
Source: Goldman Sachs Global Investment Research, Census Bureau, US Bureau of Labor Statistics

2. Technological Displacement Has Long-Lasting Negative Effects

The labor market effects of technological displacement are persistent. We estimate these effects over the ten years following a job loss by matching technology-displaced workers with comparable workers who lost their jobs in non-disrupted occupations and with never-displaced workers[3] via a propensity score matching approach—with workers paired according to their pre-displacement characteristics[4]—and computing the average gap in outcomes across matched pairs.

As the left panel of Exhibit 6 shows, real earnings for technology-displaced workers decline more sharply on impact and never fully recover, lagging those of never-displaced workers by slightly under 10pp a decade after displacement and those of other displaced workers by roughly 5pp. The probability of experiencing a subsequent unemployment spell also remains persistently higher, particularly in the first five years (Exhibit 6, right panel).

Exhibit 6: Technology-Displaced Workers Experience Persistently Lower Earnings and Greater Job Instability in the Years Following Displacement



Source: Goldman Sachs Global Investment Research, Census Bureau, US Bureau of Labor Statistics

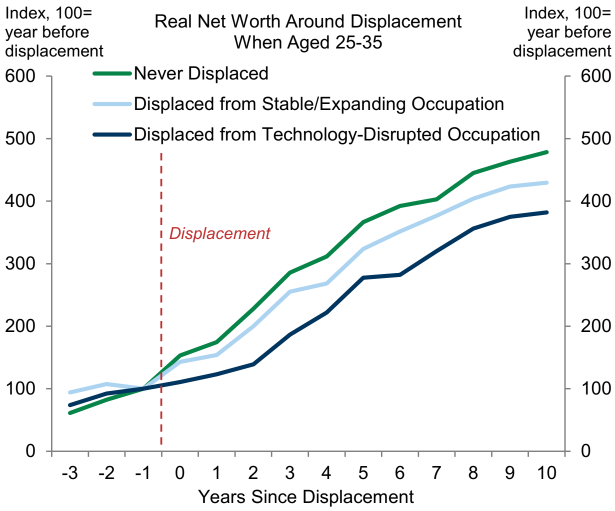
The scarring effects extend beyond labor market outcomes. Focusing on workers displaced early in their careers—between the ages of 25 and 35—we find that technological displacement meaningfully slows wealth accumulation relative to other workers (Exhibit 7, left panel). This effect is largely driven by delayed homeownership, as lower earnings and persistent job instability constrain displaced workers' ability to save and invest.

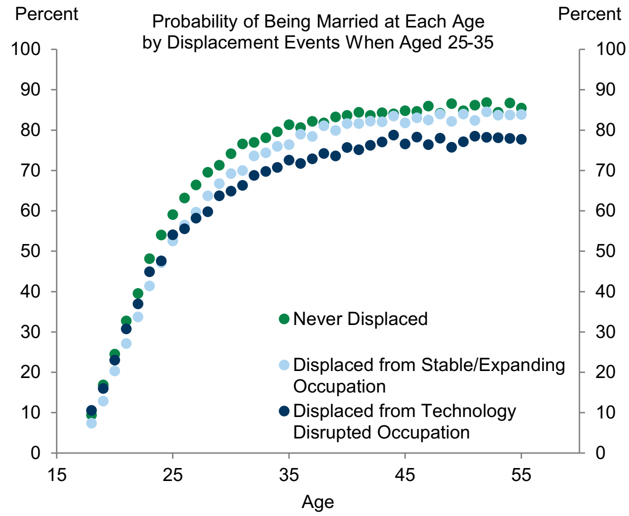
These dynamics are also reflected in delayed household formation. The probability of being married at any given age is lower for workers displaced early in their careers relative to never-displaced workers, and especially so for those displaced by technology (Exhibit 7, right panel).

[3] Non-displaced workers are workers who remain continuously employed throughout the ten-year estimation window.

[4] The matching procedure pairs workers on their demographic characteristics (age, sex, race, marital status, education level, income, and net wealth), as well as their industry, occupation tenure, year of displacement, and urban/rural location.

Exhibit 7: Lower Reemployment Earnings and Persistent Job Instability Weigh on Wealth Accumulation and Household Formation for Workers Displaced by Technology Early in Their Careers



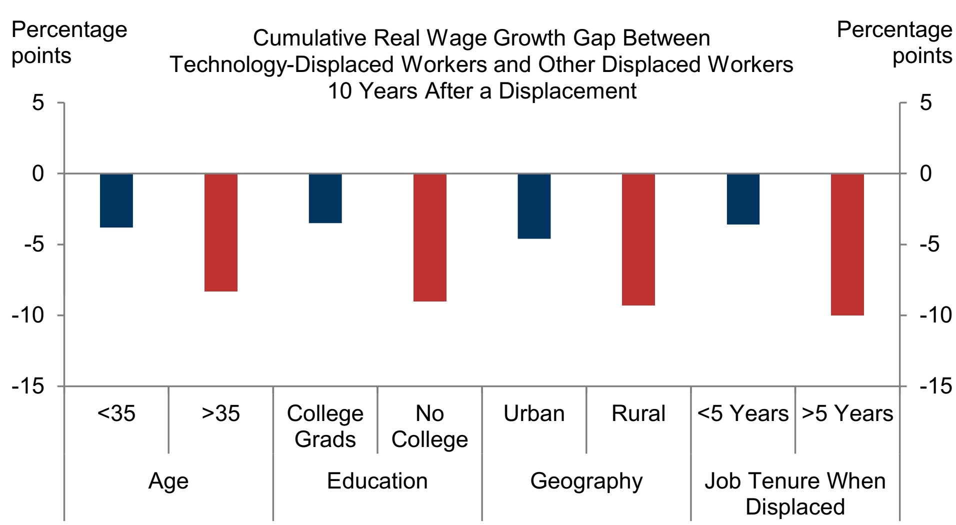


Source: Goldman Sachs Global Investment Research, Census Bureau, US Bureau of Labor Statistics

3. The Effects of Technological Displacement Are Widely Heterogeneous Across Workers

The impact of technological displacement varies considerably across workers. As Exhibit 8 shows, younger, college-educated, and urban workers experience excess cumulative real earnings losses in the first ten years after displacement roughly half as large as those of other technology-displaced workers. The key mechanism is occupational mobility. Workers in these groups are more likely to switch occupations upon displacement and to transition into roles with lower routine content—reducing their exposure to further displacement risk. Workers with shorter tenure at the time of displacement also fare better, likely because their skills are less occupation-specific and therefore more readily transferable.

Exhibit 8: Younger, More Educated, and Shorter-Tenure Workers Experience Smaller Earnings Losses Following Technological Displacement

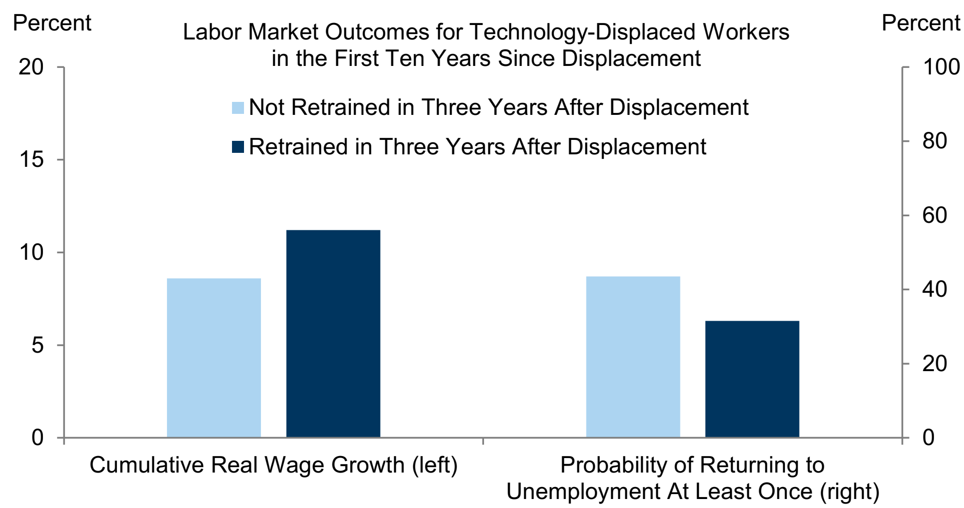


Source: Goldman Sachs Global Investment Research, Census Bureau, US Bureau of Labor Statistics

Retraining also helps to cushion the impact of displacement. The National Longitudinal Surveys provide information on whether respondents participated in vocational or technical programs, apprenticeships, government-sponsored initiatives, or employer-provided training between survey waves. To estimate the effect of these programs, we again apply a propensity score matching approach, comparing technology-displaced workers who spent at least four weeks in retraining within three years of a job loss with otherwise comparable technology-displaced workers who did not participate.

We find that retrained workers achieve roughly 2pp more cumulative real wage growth over the decade following displacement, and their probability of returning to unemployment at least once during that period declines by about 10pp (Exhibit 9).

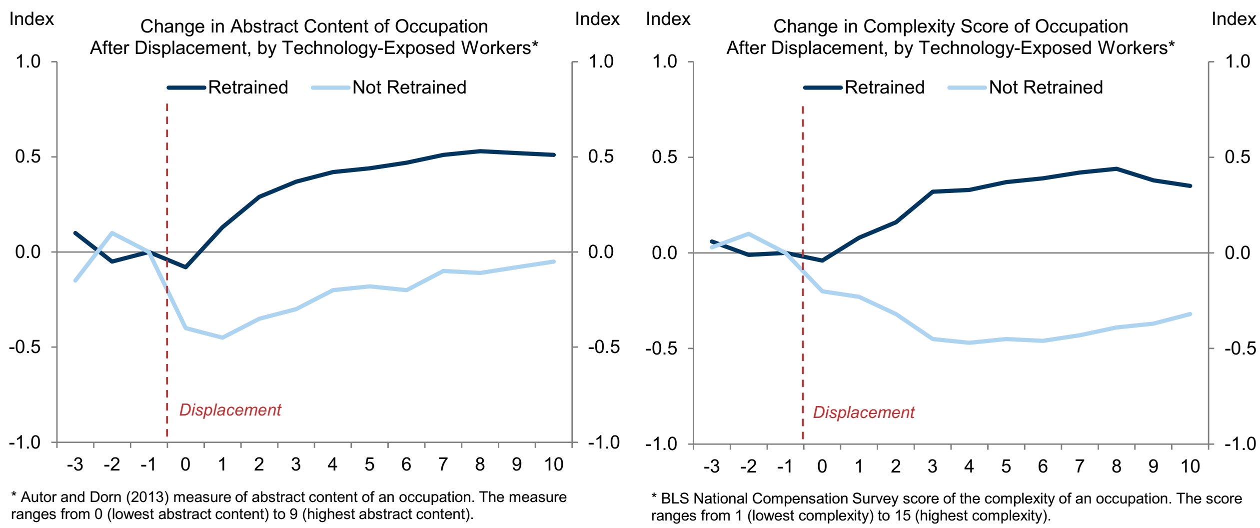
Exhibit 9: Retraining Modestly Improves the Labor Market Outcomes of Technology-Displaced Workers...



Source: Goldman Sachs Global Investment Research, Census Bureau, US Bureau of Labor Statistics

These gains reflect skill upgrading relative to other technology-displaced workers. Retrained workers tend to move up the occupational ladder into roles with higher abstract content—positions requiring advanced skills and greater complementarity with information and communication technology—thereby reducing their exposure to future automation (Exhibit 10, left panel). They also tend to transition into roles with higher complexity scores as measured by the BLS's National Compensation Survey (Exhibit 10, right panel).

Exhibit 10: ...Likely by Facilitating Transitions into More Abstract and Higher-Complexity Occupations

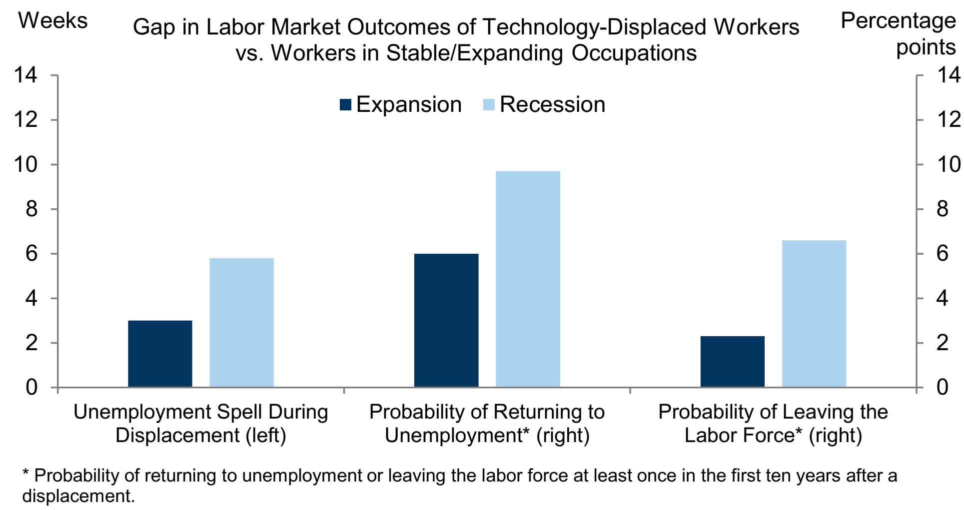


Source: Goldman Sachs Global Investment Research, Census Bureau, US Bureau of Labor Statistics

4. The Scarring Effects of Technological Displacement Are Larger in Recessions
Recessions significantly amplify the costs of technological displacement for affected workers. As we noted in previous reports, firms disproportionately shed routine jobs during downturns, when they are under greater pressure to identify efficiencies and streamline their workforce.

For individual workers, technology-induced displacement during a recession amplifies the labor market penalty already faced relative to workers displaced from other occupations. The baseline gaps estimated during expansions widen further during recessions—by about three weeks for the duration of unemployment spells and 5pp each for the risk of subsequent unemployment and the probability of exiting the labor force entirely (Exhibit 11). This likely reflects the fact that reemployment becomes harder and job matches less durable when many displaced workers compete for a limited number of vacancies in occupations that are rapidly contracting.

Exhibit 11: The Scarring Effects of Technological Displacement Are Significantly Larger During Recessions



Source: Goldman Sachs Global Investment Research, Census Bureau, US Bureau of Labor Statistics

Lessons for AI-driven Occupational Displacement

Our analysis suggests that, similarly to previous waves of technological change, AI-driven displacement could impose lasting costs on affected workers, worsening labor market outcomes for several years. These effects could be substantially larger when displacement coincides with a recession.

However, even if AI-driven job losses disproportionately affect new graduates, our findings suggest that younger workers will likely find it easier to adjust by switching jobs and upgrading skills. And encouragingly, our analysis suggests that retraining programs could help to mitigate some of the negative effects of AI-related job displacement, enabling displaced workers to earn modestly higher wages and achieve more stable employment.

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The US Economic and Financial Outlook

THE US ECONOMIC AND FINANCIAL OUTLOOK
(% change on previous period, annualized, except where noted)

						2025				2026			
	2023	2024	2025	2026	2027	Q1	Q2	Q3	Q4	Q1	Q2	Q3	Q4
OUTPUT AND SPENDING													
Real GDP	**2.9**	**2.8**	**2.1**	*2.3*	*2.0*	**-0.6**	**3.8**	**4.4**	**0.7**	*2.9*	*2.0*	*1.7*	*1.7*
Real GDP (annual=Q4/Q4, quarterly=yoy)	**3.4**	**2.4**	**2.0**	*2.1*	*2.2*	**2.0**	**2.1**	**2.3**	**2.0**	*2.9*	*2.5*	*1.8*	*2.1*
Consumer Expenditures	**2.6**	**2.9**	**2.6**	*1.8*	*1.8*	**0.6**	**2.5**	**3.5**	**2.0**	*1.5*	*1.2*	*1.4*	*1.5*
Residential Fixed Investment	**-7.8**	**3.2**	**-2.1**	*-2.2*	*1.9*	**-1.0**	**-5.1**	**-7.1**	**-0.6**	*-5.7*	*1.5*	*1.5*	*1.5*
Business Fixed Investment	**7.3**	**2.9**	**4.1**	*5.0*	*4.7*	**9.5**	**7.3**	**3.2**	**2.3**	*7.3*	*5.4*	*5.1*	*4.2*
Structures	**16.7**	**1.1**	**-5.3**	*-0.9*	*3.8*	**-3.1**	**-7.5**	**-5.0**	**-7.1**	*0.4*	*4.0*	*4.0*	*4.0*
Equipment	**2.9**	**3.5**	**8.3**	*7.2*	*5.1*	**21.3**	**8.5**	**5.2**	**3.9**	*12.4*	*6.5*	*6.0*	*4.5*
Intellectual Property Products	**6.2**	**3.5**	**5.7**	*5.9*	*4.8*	**6.5**	**15.0**	**5.6**	**5.6**	*6.0*	*5.0*	*4.8*	*4.0*
Federal Government	**3.3**	**3.8**	**-1.2**	*1.8*	*0.9*	**-5.6**	**-5.3**	**2.7**	**-16.7**	*22.5*	*0.0*	*1.0*	*1.0*
State & Local Government	**3.6**	**3.8**	**2.5**	*0.9*	*0.9*	**1.9**	**3.1**	**2.0**	**1.2**	*0.3*	*0.3*	*0.5*	*1.0*
Net Exports ($bn, '17)	**-925**	**-1,033**	**-1,091**	*-1,017*	*-1,076*	**-1,381**	**-1,058**	**-956**	**-968**	*-992*	*-1,013*	*-1,024*	*-1,039*
Inventory Investment ($bn, '17)	**47**	**44**	**31**	*18*	*41*	**172**	**-18**	**-24**	**-8**	*-9*	*25*	*25*	*30*
Nominal GDP	**6.7**	**5.3**	**5.0**	*6.2*	*4.1*	**2.9**	**6.0**	**8.3**	**4.5**	*7.6*	*7.7*	*3.0*	*3.6*
Industrial Production, Mfg.	**-1.0**	**-1.0**	**0.9**	*2.6*	*3.2*	**4.0**	**2.5**	**2.5**	**-2.4**	*5.3*	*4.1*	*3.2*	*2.9*
HOUSING MARKET													
Housing Starts (units, thous)	**1,421**	**1,371**	**1,357**	*1,286*	*1,322*	**1,401**	**1,354**	***1,346***	**1,328**	*1,280*	*1,290*	*1,280*	*1,292*
New Home Sales (units, thous)	**665**	**685**	**672**	*670*	*644*	**655**	**665**	***688***	**680**	*690*	*690*	*662*	*638*
Existing Home Sales (units, thous)	**4,103**	**4,067**	**4,076**	*4,212*	*4,375*	**4,087**	**4,013**	**4,047**	**4,157**	*4,169*	*4,189*	*4,223*	*4,267*
Case-Shiller Home Prices (%yoy)*	**5.3**	**3.8**	**0.6**	*0.7*	*2.2*	**3.9**	**2.4**	**1.5**	**0.6**	*0.3*	*0.8*	*0.7*	*0.7*
INFLATION (% ch, yr/yr)													
Consumer Price Index (CPI)**	**3.3**	**2.9**	**2.7**	*3.4*	*2.2*	**2.7**	**2.5**	**2.9**	**2.7**	*3.0*	*4.1*	*3.4*	*3.4*
Core CPI **	**3.9**	**3.2**	**2.6**	*2.3*	*2.2*	**3.1**	**2.8**	**3.1**	**2.7**	*2.5*	*2.7*	*2.4*	*2.3*
Core PCE** †	**3.1**	**3.0**	**3.0**	*2.5*	*2.0*	**2.8**	**2.7**	**2.9**	**2.9**	*3.0*	*3.0*	*2.9*	*2.7*
LABOR MARKET													
Unemployment Rate (%)^	**3.8**	**4.1**	**4.4**	*4.6*	*4.5*	**4.2**	**4.1**	**4.4**	**4.4**	**4.3**	*4.4*	*4.5*	*4.6*
U6 Underemployment Rate (%)^	**7.2**	**7.6**	**8.4**	*8.7*	*8.4*	**7.9**	**7.7**	**8.1**	**8.4**	**8.0**	*8.2*	*8.5*	*8.7*
Payrolls (thous, monthly rate)	**210**	**122**	**10**	*46*	*92*	**20**	**34**	**23**	**-39**	**68**	*35*	*35*	*47*
Employment-Population Ratio (%)^	**60.1**	**59.9**	**59.7**	*58.9*	*58.8*	**59.9**	**59.7**	**59.7**	**59.7**	**59.2**	*59.1*	*59.0*	*58.9*
Labor Force Participation Rate (%)^	**62.5**	**62.5**	**62.4**	*61.8*	*61.6*	**62.5**	**62.3**	**62.5**	**62.4**	**61.9**	*61.9*	*61.8*	*61.8*
Average Hourly Earnings (%yoy)	**4.2**	**5.4**	**4.6**	*4.0*	*3.9*	**5.0**	**4.6**	**4.5**	**4.4**	**4.2**	*4.1*	*3.9*	*3.8*
GOVERNMENT FINANCE													
Federal Budget (FY, $bn)	**-1,694**	**-1,833**	**-1,775**	*-2,050*	*-2,200*	--	--	--	--	--	--	--	--
FINANCIAL INDICATORS													
FF Target Range (Bottom-Top, %)^	**5.25-5.5**	**4.25-4.5**	**3.5-3.75**	*3-3.25*	*3-3.25*	**4.25-4.5**	**4.25-4.5**	**4-4.25**	**3.5-3.75**	**3.5-3.75**	*3.5-3.75*	*3.25-3.5*	*3-3.25*
10-Year Treasury Note^	**3.88**	**4.58**	**4.18**	*4.20*	*4.25*	**4.23**	**4.24**	**4.16**	**4.18**	**4.30**	*4.20*	*4.20*	*4.20*
Euro (€/$)^	**1.11**	**1.04**	**1.17**	*1.19*	*1.20*	**1.08**	**1.18**	**1.17**	**1.17**	**1.15**	*1.14*	*1.18*	*1.19*
Yen ($/¥)^	**141**	**157**	**157**	*155*	*141*	**150**	**144**	**148**	**157**	**160**	*160*	*158*	*155*

* Weighted average of metro-level HPIs for 381 metro cities where the weights are dollar values of housing stock reported in the American Community Survey. Annual numbers are Q4/Q4.
** Annual inflation numbers are December year-on-year values. Quarterly values are Q4/Q4.
† PCE = Personal consumption expenditures. ^ Denotes end of period.
Note: Published figures in bold.
Source: Goldman Sachs Global Investment Research.

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Disclosure Appendix

Reg AC

We, Pierfrancesco Mei and Jessica Rindels, hereby certify that all of the views expressed in this report accurately reflect our personal views, which have not been influenced by considerations of the firm's business or client relationships.

Unless otherwise stated, the individuals listed on the cover page of this report are analysts in Goldman Sachs' Global Investment Research division.

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Disclosures

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